Contrastive Learning Based on Multiscale Hard Features for Remote-Sensing Image Scene Classification

Zhihao Li, Biao Hou, *Member, IEEE*, Xianpeng Guo, *Student Member, IEEE*, Siteng Ma, Yanyu Cui, Shuang Wang, *Senior Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— The overwhelming majority of models for remote-sensing image (RSI) scene classification generally require the weights pretrained on natural images for initialization before formal training. However, differences in imaging mechanisms lead to huge discrepancies between natural images and RSIs, and the strong visual representation learned from massive natural images limits the performance of models when inferencing RSIs. To address this issue, the well-established self-supervised contrastive learning paradigm in the natural image field is introduced to the RSI field. We propose a contrastive learning method based on multiscale hard features (MHCL), which aims to use finite RSIs to learn sufficient visual representations in an unsupervised contrastive manner, thus providing a powerful upstream pretrained model for fine-tuning downstream scene classification task. Multilevel features extracted by intermediate layers of each encoder's backbone are first gathered and then a hard features transformation method (HFT) is proposed to create hard positive features and diverse queues that save hard negatives, thereby enriching the finite scene information in small-scale RSIs. Furthermore, we redesign the multiscale hard features joint contrastive loss to boost the model to explore sufficient invariant representations by additionally pulling hard positive pairs closer and pushing hard negative pairs farther away in the embedding space. Extensive experiments demonstrate that the upstream pretraining model generated by MHCL achieves competitive transferred performance on three popular scene classification datasets, outperforming the traditional model pretrained on ImageNet and models pretrained by other state-of-the-art contrastive learning methods. Our code will be released at: https://github.com/benesakitam/MHCL.**

***Index Terms*— Hard feature transformation, model pretraining, remote-sensing image (RSI) scene classification, self-supervised contrastive learning.**

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Zhihao Li and Siteng Ma are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China.

Biao Hou, Xianpeng Guo, and Yanyu Cui are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education of China, Xidian University, Xi'an 710071, China (e-mail: avcodec@hotmail.com).

Shuang Wang is with the Institute of Intelligent Information Processing, Xidian University, Xi'an 710071, China.

Licheng Jiao is with the School of Electronic Engineering, Xidian University, Xi'an 710071, China.

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I. Introduction

REMOTE-SENSING image (RSI) scene classification, as a basic application tool in remote sensing, plays a pivotal role in modern geological prospecting [1], urban planning [2], precision agriculture [3], disaster prevention [4], vector-borne disease surveillance [5], tackling climate change [6], and other fields involving national defense and people's well-being.

Deep learning has been widely used in RSI scene classification [7], [8], [9], [10] and has made significant progress. However, it should be noted that previous works have often required a supervised pretrained model as initialization. The generally used pretrained model produced by representation learning on the natural image dataset ImageNet [11] has considerable limitations in fine-tuning the downstream RSI scene classification task. This mainly stems from the huge discrepancy in the data types and imaging mechanisms between natural images and RSIs. However, the great development of technology has produced various RSIs with higher resolution (spatial resolution and spectral resolution) and quality and refined annotations usually require expert knowledge or expensive ground sensors. It is obviously impractical to construct an RSI dataset at the same scale as ImageNet, which contains more than 10 million labeled images in actual applications. Several recent methods [12], [13] have been developed to alleviate the requirement for labeled data, but there are still many difficulties in landing on the application of RSIs. The performances of many interesting downstream tasks are limited due to the lack of reasonable means of initialization. With the ever-increasing appearance of observation satellites in orbit, terabytes of imagery data are generated every day. Therefore, timely and sufficient analysis and understanding of these data without annotations has become an increasingly urgent demand for civil and military use.

Recently, self-supervised representation learning [14], [15], [16] in the natural image field has facilitated the advancement of many downstream tasks, such as image classification [11], object detection [17], [18], and semantic segmentation [19]. Among them, benefiting from the unique paradigm of pulling the representation of positive pairs (i.e., views of the same

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image) closer while pushing negative pairs (i.e., views of different images) farther away, contrastive learning [14], [20], [21], [22], [23], [24] has attracted wide attention, which can learn sufficient invariant visual representations from unlabeled images, thus providing a general upstream pretrained model for downstream tasks without labels. We hope to introduce contrastive learning to provide a more appropriate means of initialization for the basic scene classification task. Unfortunately, it is unreasonable to directly deploy mature contrastive learning methods to treat RSIs. Compared with natural images, RSIs cover complex scenes, and the distribution of object size as the basis for classification is uneven. Moreover, differences in satellite quality also have adverse influences on imaging, such as visual blur and image distortion. In addition, the scale of currently released RSI datasets is far less than that of many published natural image datasets. To overcome the problems above, we propose a new **c**ontrastive **l**earning method based on **m**ultiscale **h**ard features (MHCL), which can create the discriminative multiscale hard features from small-scale RSIs only, thus exploring sufficient invariant visual representations.

Specifically, we first feed two views of each RSI generated by data augmentation into dual-encoders. The intermediate layers of each encoder's backbone can extract shallow texture and deep semantic information at different levels of the view. Therefore, we gather the extracted multilevel intermediate features and deploy the multilevel **i**ntermediate **f**eatures **c**ontrastive **l**earning (IFCL) to mine comprehensive information. Meanwhile, a multilevel contrastive loss is designed to directly optimize multilevel features, ensuring the model has great discrimination when distinguishing objects with different sizes in complex scenes. Next, owing to the lack of scene information contained in small-scale RSI datasets, we propose a **h**ard **f**eatures **t**ransformation method (HFT) to transform the easy multilevel features extracted directly into diverse hard positive counterparts and negatives in the embedding space. HFT can fuse the real information contained in various views to break through the limitation of the inherent scene information in datasets, creating extra synthetic scene information. Finally, we extend the multilevel contrastive loss to a multiscale hard feature joint contrastive loss, which not only learns sufficient invariant representations from the finite scene information of multilevel easy features, but also boosts the discrimination of the model with the help of synthetic scene information of hard features, Therefore, MHCL can provide a reasonable common upstream pretrained model for fine-tuning the downstream scene classification task better.

The contributions of this article are thus summarized as follows.

1) This article proposes a novel MHCL for the scene classification task. We aim to contribute a general upstream means of initialization to the RSI community from the pretraining perspective.
2) We deploy the multilevel IFCL to learn more comprehensive invariant information from intermediate features by the designed multilevel contrastive loss, which also enhances the discrimination of the model to distinguish objects with uneven sizes in complex scenes.
3) We propose an HFT. The additional hard features created by HFT contain the synthetic scene information, which enriches the scene information inherent in finite RSIs.
4) We extend the multilevel contrastive loss to a multiscale hard features joint contrastive loss, which can further boost the model to learn more sufficient invariant representations from multilevel easy features and hard ones.
5) Extensive experimental results demonstrate the effectiveness and superiority of MHCL on three popular released scene classification datasets: UC Merced, NWPU-RESISC45, and MLRSNet.

II. Related Works

In this section, we briefly discuss some related works about RSI scene classification and self-supervised contrastive learning.

A. Remote-Sensing Image Scene Classification

RSIs usually cover complex scenes, each scene may contain various land-cover units with different sizes and appearances. Therefore, mining discriminative representations that can sufficiently describe scene information becomes the key to the RSI scene classification task. Traditional methods rely on handcrafted descriptors such as scale-invariant feature transform (SIFT) [25], histogram of oriented gradients (HOG) [26], and the object spectral index [27] to represent remote-sensing scenes. Furthermore, some works introduce the visual bag-of-visual-words model [28], [29], probabilistic topic model (PTMs) [30], [31], vectors of locally aggregated descriptors (VLADs) [32], and the Fisher kernel [33] to extract local attributes of RSIs and encode visual representations. However, the above scene classification methods only focus on relatively simple lower-level features and cannot adequately capture the higher-level semantic information contained in remote-sensing scenes.

Nowadays, deep learning has achieved widespread success in the natural image field, where the convolutional neural network (CNN) has received wide attention from researchers due to its powerful feature extraction ability. In the RSI field, a series of CNN-based methods [7], [8], [9], [10], [34], [35], [36], [37], [38] have also been proposed and achieved significant breakthroughs in scene classification tasks. However, these approaches first still require the model pretrained on the very large-scale natural image dataset ImageNet [11] to initialize the backbone, followed by fine-tuning the scene classification dataset to learn the scene representations. To overcome the inherent discrepancies between RSIs and natural images during fine-tuning, several works assemble CNNs with different structures [39], [40] or propose diverse feature aggregation strategies [35], [41], [42] for feature learning. In addition, there are methods to drive the model to adapt upstream visual representations from nature images by designing unique network structures [9], [10], [34], setting up the two-stream branches [36], [43], extracting frequency and spatial domain representations [44], designing the new discriminative loss function [45], or introducing the domain adaptation network [46]. However, the large investment during fine-tuning of the above works limits the real potential of the proposed models. In this article, we start with the pretraining

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perspective and provide an elegant and powerful common upstream pretrained model specifically for the RSI field by introducing self-supervised contrastive learning.

B. Self-Supervised Contrastive Learning

Considering the large amount of unlabeled data that exists in the real world and the expensive cost of labeling work, machine-learning researchers have turned their attention to developing self-supervised representation learning methods recently. Generally, the success of a self-supervised learning algorithm [14], [15], [16], [47] depends on two aspects: the pretext task and loss function. Self-supervised learning methods train a predictive model without labels from the input signal by designing various pretext tasks. Image colorization [48], spatial transformation [49], [50], the image jigsaw puzzle [51], relative patch prediction [52], [53], image super-resolution [54], and image painting [55] are useful types of representation learning. Simultaneously, the pretext task requires loss functions of some form to better investigate the more robust representation. A recent effective way replaces a dense per-pixel predictive loss with a loss in a lower-dimensional representation space, inspired by noise contrastive estimation [56], [57] or N-pair losses [58]. The state-of-the-art family of models for self-supervised representation learning using this paradigm are assembled under the umbrella of contrastive learning [20], [21], [22], [23], [24], [56], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70]. Some of these methods [21], [60], [61], [63], [64] only apply the contrastive loss to the last layer of a deep network. During testing, the embeddings from a previous layer are utilized for downstream fine-tuning, transfer, or direct retrieval tasks.

The core idea of contrastive learning is to bring the positive sample pairs closer and repulse the negative sample pairs [20], [21], [47]. Chen et al. [21] designed a simple contrastive framework, SimCLR, for learning visual representation from the augmented views of other negative samples in a minibatch. He et al. [20] proposed MoCo, which uses a momentum-updated memory bank of old negative items to maintain consistent representations from large batches of negative samples. After that, with the deployment of multilayer perceptron (MLP) and a more effective augmented version of views, MoCo v2 [66] was developed as an important baseline for motivating progress in contrastive learning research. Grill et al. [22] proposed the BYOL method, which leverages positive pairs alone for iteratively bootstrapping the model to learn the enhanced representation. Furthermore, SimSiam [71], a variant of BYOL, was proposed to eliminate the requirement for negative pairs by using diverse augmented views of the same sample.

A large proportion of contrastive methods are unsupervised, but there are still some methods [72], [73] that use the label information by constructing the new supervised contrastive loss function. In addition, cluster algorithms have also been successfully applied in unsupervised contrastive learning by alternating cluster representations and learning to predict cluster assignments. Caron et al. [70] treated the cluster assignments as pseudo-labels to optimize the classification objective. PCL [74] was proposed to optimize the contrastive loss on prototypes (i.e., cluster centers), which enables representation learning to occur directly in the lower-dimensional embedding space. Concurrently, SwAV [23] introduced an online clustering strategy and used a large memory bank to provide abundant samples for clustering without computing pairwise comparisons. Of course, the research on contrastive learning has also covered other aspects: [75], [76] performing pretext tasks such as object detection, mining hard negative samples [77], [78] from the perspective of noise contrastive estimation loss, and providing more efficient data augmentation strategies [79], [80], [81] to facilitate contrastive learning.

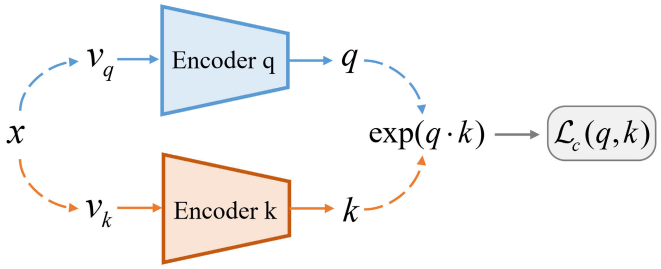


Fig. 1. Common paradigm of contrastive learning. Contrastive loss calculates the similarity score between representations of different augmentation views generated from images to push each negative pair far away and pull positive pair closer.

Currently, self-supervised contrastive learning is deployed in various research based on natural images, but it is rarely seen in the remote-sensing field, which is regrettable. Therefore, we propose a new contrastive learning method, MHCL, which is good at treating RSIs. We hope to contribute a powerful upstream means of initialization to the RSI community for meeting the demand for basic downstream tasks.

III. Method

Some magnum opuses such as MoCo v1/v2 [20], [66] have made breakthroughs in the research of pretraining on natural images. We draw the essential paradigm of this classic contrastive learning to construct a baseline and further propose MHCL.

A. Paradigm

The core step of contrastive learning is shown in Fig. 1. For each image x without a label, two separated data augmentation operations, which are randomly sampled from the same data augmentation pool, can generate two different views. Afterward, the query and key views (v_q and v_k, respectively) are fed into two encoders with the same structure to extract features. Each encoder consists of the backbone network connected to a projection head, and we only transfer the model of the backbone and discard the projection head after pretraining. Consider an encoded query q and a set of encoded keys $\{k^0, k^1, k^2, \dots\}$ saved in the memory bank. There must exist one single positive key k_+ that matches the query q because the encoded key and query are generated from different views of the same image. Then, q and k_+ are constructed as a positive pair [21], [22], and we employ the contrastive loss to pull

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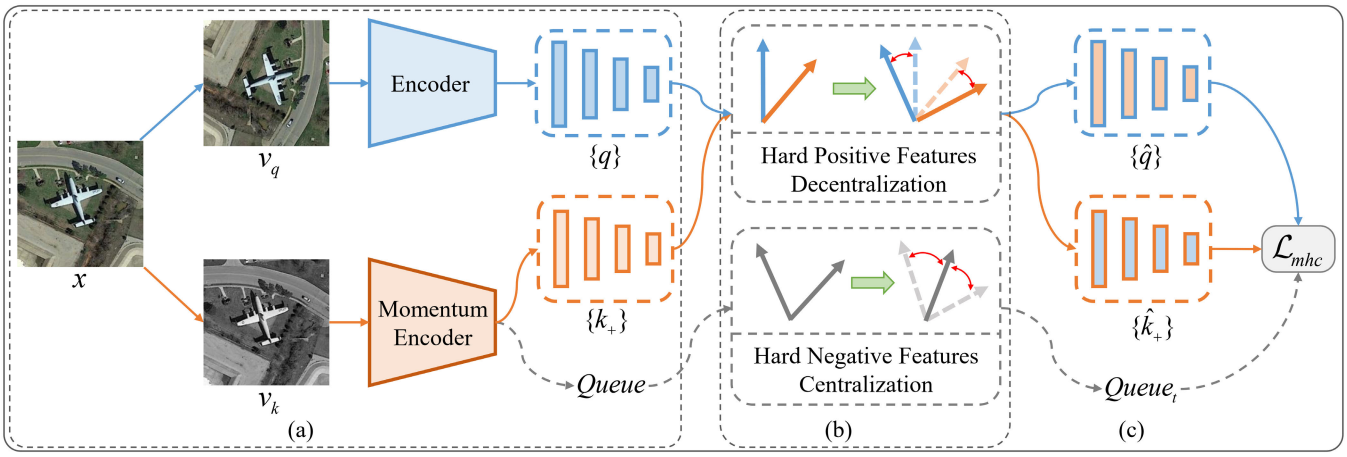


Fig. 2. (a) Feature extraction of multilevel IFCL, which gathers the multilevel features encoded by each intermediate layer of the backbone of two encoders. (b) Proposed HFT. We transform all easy multilevel features obtained from (a) into hard positive or negative ones to create the extra scene information. (c) Overview of MHCL. We propose the multiscale hard features joint contrastive loss $\mathcal{L}_{\text{mhc}}$ to fully leverage the multilevel easy and hard features for representation learning better. Note that Queue$_t$ expresses the diversity queues after transformation.

closer the positive pair and increase the distance between the query and other all negative keys, termed k_-. The **c**ontrastive loss function here generally adopts infoNCE [62] loss, which is described in this article as

$$\mathcal{L}_c(q,k) = -\log \frac{\exp(q \cdot k_+/\tau)}{\exp(q \cdot k_+/\tau) + \sum_{k_-} \exp(q \cdot k_-/\tau)} \quad (1)$$

where τ is the temperature hyperparameter [64]. We first construct a **baseline** model with the dual-encoder and memory bank structure following the crucial contrastive paradigm described above (see Section IV-B for detailed settings).

B. Overview

A novel MHCL is proposed in this article. The overview of MHCL is shown in Fig. 2, where each encoder contains multiple projection heads instead of a single projection head in the baseline's encoder. First, we feed the augmented views of RSIs without labels into encoders. The feature maps extracted by intermediate layers of the backbone are sent into the corresponding projection head with different levels, respectively. We obtain multiple intermediate feature vector groups containing different levels of abundant information of views. Thus, the multilevel IFCL is deployed to comprehensively learn invariant information from multilevel features for enhancing the discrimination of the model. Then, we propose an HFT, which performs hard positive features decentralization and hard negative features centralization operations on each feature vector group of the same level. With the fusion and swap of each group in the embedding space, the hard features generate not only maintain primitive real-scene information, but also create new synthetic information in a data augmentation manner. Ultimately, we redesign multiscale hard features joint contrastive loss to learn more invariant representation from multilevel easy and hard features, which can also boost the discrimination of the model. During the evaluation, we transfer the MHCL pretrained model as the backbone to fine-tune the downstream scene classification task.

C. Multilevel Intermediate Features Contrastive Learning

Considering the challenges of scene complexity and unequal scale in RSIs, we follow the concepts of momentum update strategy and dynamic queue from MoCo v2 [66]. Simultaneously, a multilevel contrastive loss is designed to fully leverage the representations encoded from intermediate layers of the backbone to boost the model and distinguish diverse scenes more discriminative.

As shown in Fig. 2(a), the views v_q and v_k drawn by the given unlabeled images $x \in \mathbb{R}^{h\times w\times 3}$ are fed into the encoders E_q and E_k^m, respectively. The two encoders have the same architecture; both utilize ResNet-50 [82] as the backbone network, where the weights of the backbone in E_k^m are smoothly endowed from the backbone in E_q through the momentum update mechanism [66]. We gather the shallow texture and deep semantic representations output from the intermediate layer2, layer3, layer4, and layer5 of the backbone and sent these to four projection heads with different input dimensions to generate the easy multilevel features. Actually, the projection head is an MLP consisting of two linear layers and one ReLU activation layer, and the weights of four MLPs are not shared during training. As a result, we can obtain four groups of query vectors via $\{q^2, q^3, q^4, q^5\} = E_q(v_q)$; similarly, the results of key views can be obtained by $\{k^2, k^3, k^4, k^5\} = E_k^m(v_k)$. To comprehensively learn invariant information of multilevel features for enhancing the discrimination of the model, we extend the classic infoNCE loss toward a **m**ulti**l**evel **c**ontrastive loss function

$$\mathcal{L}_{\text{mlc}}\left(q^i, k^i\right) = -\sum_{i=2}^{5} \omega^i \cdot \log \frac{\exp\left(q^i \cdot k_+^i/\tau\right)}{\exp\left(q^i \cdot k_+^i/\tau\right) + \sum_{k_-} \exp\left(q^i \cdot k_-^i/\tau\right)} \quad (2)$$

where i represents the current layer of the backbone network. ω^i is the weight of the loss function, and when calculating the loss for shallow features, we set a lower weight inspired by

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PSPNet [83]; vice versa, a larger weight is assigned for deep representations. Note that for each group of features of the same scale $\{q^i, k^i\}$, we construct an independent queue [66] as a memory bank to save all negative encoded keys of $\{q^i\}$.

D. Hard Features Transformation

Given the general small-scale RSI dataset, the scene information contained in each image should be fully mined and leveraged. Some recent works [79], [80], [81] have proposed effective data augmentation strategies to enhance the understanding of models facing finite RSIs. Inspired by Lee et al. [80] and Zhu et al. [81], we note that after the multilevel features output by encoders complete through L2-normalization, all $\{k^i\}$ and $\{q^i\}$ are mapped into a unified unit spherical space. Therefore, we propose an HFT within each group of feature vectors, including hard positive features decentralization and hard negative features centralization operations. In this way, the directly encoded easy queries and keys incorporate each other to additionally transform into hard counterparts. The synthetic scene information contained in generated hard features can lift the limitation of the inherent information in the dataset, providing a strong foundation for the model to explore more abundant discriminative information.

1) Hard Positive Features Decentralization: From the perspective of maximizing the use of images, the direct idea is to obtain more features that contain discriminative information. Generally, in a standard unit sphere space, a larger angle between two normalized features from the center of the sphere to the spherical surface denotes less similarity (i.e., two features of a large angle can be viewed as hard features of each other). So, we increase the angle between each pair of queries and its positive key to create hard positive features through the decentralization operation. Specifically, as shown in Fig. 3(a), q^i are extrapolated by adopting weighted subtraction to generate the centrifugal hard features of q^i; likewise, the hard features of k_+^i are obtained. This process can be written as

$$\begin{cases} \hat{q}^i = (1+\lambda)\cdot q^i - \lambda \cdot k_+^i \\ \hat{k}_+^i = (1+\lambda)\cdot k_+^i - \lambda \cdot q^i \end{cases} \tag{3}$$

where the weights are $\lambda \sim \text{Beta}(\alpha, \beta)$ and $\lambda \in (0, 1)$. The smaller the λ, the more hard the generated features are. Owing to the small scale of RSI datasets compared to natural image datasets such as ImageNet [11], overly hard features destroy some unique scene information in the images, which is not conducive to the model prediction. Thus, we choose a larger λ to create hard positive features. The influences of different λ values are presented in Section IV-C.

2) Hard Negative Features Centralization: During each epoch of training, the memory queue in the baseline only updates the foremost batch size negative keys. This situation causes the model to calculate repeatedly the loss of the remaining negative samples that are not updated in time. Therefore, we aim to increase the similarity between different negative features to construct diverse hard negatives to reduce the cost of repeated computation. Formally, we define that each group of negative keys with the same scale is stored in a separate queue $\{k_-^i\} = \{k_1, k_2, \ldots, k_Q\}_-^i$, where Q is the size of each queue, and we set $\{k_-^i\}_{\text{rand}}$ as the random permutation version of the queue $\{k_-^i\}$. As shown in Fig. 3(b), the proposed centralization operation is deployed to reduce the angle between negatives in $\{k_-^i\}$ and random permutations in $\{k_-^i\}_{\text{rand}}$ by weighted addition

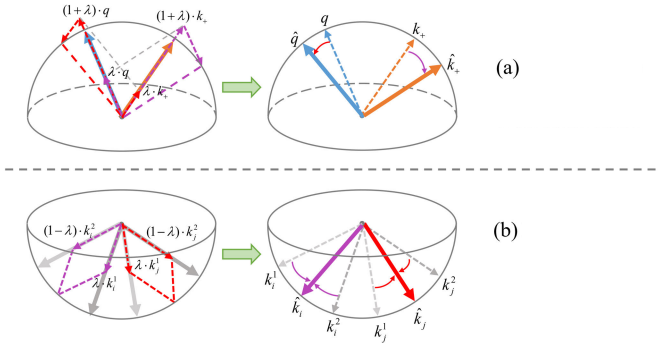


Fig. 3. Proposed HFT, including decentralization to extrapolate original queries and positive keys to generate hard positive features, as well as centralization to create hard negative features to satisfy queue diversity. The HFT converts multilevel easy representations into hard positives and negatives, which enables better IFCL in a data augmentation manner. (a) Hard positive features decentralization. (b) Hard negative features centralization.

$$\{\hat{k}_-^i\} = \lambda \cdot \{k_-^i\} + (1-\lambda)\cdot \{k_-^i\}_{\text{rand}} \tag{4}$$

where λ is the same as in hard positive features decentralization. We also assign λ a large value to avoid generating hard negatives with the information collapse described above. The centralizing queue $\{\hat{k}_-^i\}$ can provide diverse negative features to boost the multilevel contrastive learning more.

E. Multiscale Hard Features Joint Contrastive Loss

After decentralizing easy positive features to generate hard positive ones $\hat{q}^i$ and $\hat{k}_+^i$ and create memory queues $\{\hat{k}_-^i\}$ covering hard negative features, we also leverage the multilevel contrastive loss at each scale to push the hard negative pairs away and attract the hard positive pairs. Extending the multilevel contrastive loss proposed in (2), we finally redesign the whole **m**ultiscale **h**ard features joint **c**ontrastive loss function to guide the model to learn more invariant representations

$$\mathcal{L}_{\text{mhc}}(q, k) = \mathcal{L}_{\text{mlc}}\left(q^i, k^i\right) + \omega_h \cdot \mathcal{L}_{\text{mlc}}\left(\hat{q}^i, \hat{k}^i\right) \tag{5}$$

where ω_h is the hyperparameter that controls the strength of the hard features' participation in optimizing the model. When transferring the model to the downstream scene classification task, some synthetic scene information contained in the transformed hard features confuses the discriminative ability of the model to a certain extent. This condition is more obvious on small datasets. Therefore, we set different ω_h according to the scale of datasets, and the hard features can be fully used to mine the potential of the model while resisting the risk of such misprediction.

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IV. Experiments

A. Dataset Description

A series of experiments are performed on three common benchmarks: the UC Merced dataset [84], NWPU-RESISC45 dataset [8], and MLRSNet dataset [85].

1) UC Merced Dataset: The UC Merced dataset contains 21 land-use classes, and each category has 100 images with the size of 256 × 256 pixels. There are a total of 2100 RGB images from the United States Geological Survey (USGS) National Map.

2) NWPU-RESISC45 Dataset: The NWPU-RESISC45 dataset is a popular benchmark constructed by Northwestern Polytechnical University, which consists of 31 500 images with RGB color space acquired from Google Earth. All images are equally divided into 45 classes, and each class has 700 images of size 256 × 256 pixels.

3) MLRSNet Dataset: The MLRSNet dataset is a currently proposed large-scale high-resolution multilabel remote-sensing dataset. MLRSNet is also extracted from Google Earth and contains 109 161 images that are annotated into 46 broad categories. The number of images in each broad class varies from 1500 to 3000. The images have a fixed size of 256 × 256 pixels with various pixel resolutions. Moreover, each image in broad classes has different subtags varying from 1 to 13.

In our experiments, all images of each category from each dataset are randomly divided into two parts. One part is used for training, including unsupervised generating a pretrained model and fine-tuning the pretrained model on the scene classification task. The other part is used for testing. The UC Merced dataset is randomly divided into 60% for training and 40% for testing. For the NWPU-RESISC45 dataset, we randomly sample 20% and 80% of the total images for training and use the remaining 80% and 20% for testing. These two split datasets are called NWPU 2-8 and NWPU 8-2, respectively. Likewise, the MLRSNet dataset, according to [85], is split into 40% and 60% for training and testing, respectively, and we only use the broad category labels in the evaluation stage.

B. Experimental Settings

1) Pretraining Details: During pretraining, all training images are first resized to 224 × 224 by random cropping and undergo one random operation from the data augmentation pool containing random color jittering, random horizontal flip, random grayscale conversion, blur augmentation [21], and color distortion [21] to generate new views. We set the initial learning rate is 0.015 and use a cosine learning rate scheduler. The SGD optimizer is used to train the model, and its weight decay and momentum are set at 0.0001 and 0.9, respectively. We set the batchsize is 128 and train 800 epochs to obtain a competitive pretrained model. Following MoCo v2, the size of queue Q is 65 536 and the temperature parameter τ is 0.2. But the momentum coefficient of the momentum update mechanism is set at 0.99 instead of 0.999 to decrease the encoder parameter update speed. Note that the model trained with the above parameters is treated as the **baseline**.

We then report other hyperparameters of the MHCL method. The weight of multilevel contrastive loss $\omega^i = 0.1, 0.4, 0.7,$ and 1.0 when $i = 2, 3, 4,$ and 5, respectively. We set the various transformation weight λ on different datasets, where λ follows a beta distribution, and its value is determined by α and β. Specifically, for UC Merced, $\alpha = 20$ and $\beta = 0.1$; for NWPU 2-8, $\alpha = 10$ and $\beta = 0.1$; for NWPU 8-2, $\alpha = 10$ and $\beta = 1$; and for MLRSNet, we set $\alpha = 1$ and $\beta = 0.1$. For the weight in multiscale hard features joint contrastive loss, we set $\omega_h = 1$ on MLRSNet, and ω_h is set at 0.1 on other datasets.

2) Evaluation Protocol: Following the linear classification protocol [20], the pretrained ResNet-50 model is transferred as the backbone of the RSI scene classification task. We freeze all feature layers except the last fully connected layer of the transferred backbone. All pretraining features to fine-tune a simple linear classifier (the last fully connected layer followed by softmax) after the global average layer. In addition, the standard cross-entropy loss and SGD optimizer for which the weight decay is 0 [63] are introduced to complete the training process. We set the initial learning rate as 0.1 and batchsize as 256, and train 100 epochs. The results of top-1 accuracy on the testing sets are reported in the following experiments, and the results of top-5 accuracy and training time cost are reported additionally in the comparison experiments.

All of our experiments are based on the PyTorch platform and deployed on four NVIDIA GeForce RTX 2080Ti GPUs.

C. Selection of Parameters

We investigate the essential hyperparameters in the proposed method, including the weight λ of hard features transformation and the weight ω_h in the multiscale hard features joint contrastive loss. The value of only one hyperparameter is changed in every experiment while fixing the other, and the corresponding results evaluating on scene classification task are detailed in this section.

1) Weight λ of Hard Features Transformation: λ follows the beta distribution, and the value is jointly determined by α and β, where its mean is $\alpha/\alpha + \beta$. beta distribution provides a probability distribution of the λ value being selected. For example, as shown in Fig. 4(a), when α and β are both 1, that is, the mean value is 0.5, the probability of any $\lambda \in (0, 1)$ be taken is the same. Due to the small-scale RSI datasets, we hope to give a large value of λ for each dataset, thus creating harder features to enrich the scene information on the basis of retaining the original information as much as possible. We adjust α and β according to the mean of the beta distribution to ensure the large λ. During training, the model deploys different λ based on the special beta distribution to generate hard features with high randomness. The evaluation results of scene classification are exhibited in Fig. 4(b). We can see that larger λ values tend to achieve better classification accuracy on small- and medium-scale datasets, while for the large dataset, any λ has competitive performance.

Specifically, for the large MLRSNet in Fig. 4(b), most of the results of arbitrarily selecting α and β exhibit a classification performance exceeding 93%. The reasons are that MLRSNet contains rich scene information, and the synthetic information

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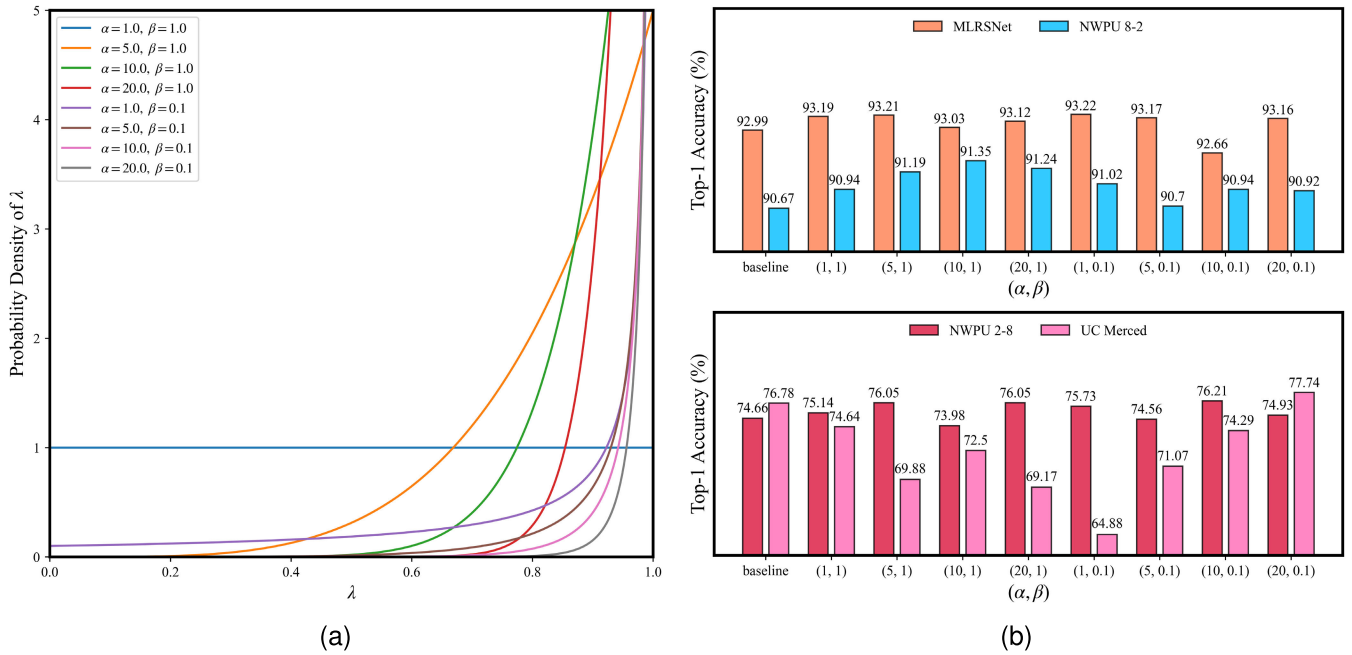


Fig. 4. (a) Visualization of the beta distribution for determining the value of transformed weight λ. (b) Classification results of leveraging the different λ in each RSI dataset.

transforming the extremely hard features (the smaller mean value of λ) not only enriches the original data, but also produces few adverse effects. In addition, tiny hard features also play a role in leak-filling for raw information. We select α and β as 1 and 0.1, respectively, to achieve the best classification accuracy of 93.22%.

In NWPU 8-2, the performance is 91.35% when $\alpha = 10$ and $\beta = 1$; at this point, λ has a high probability of being greater than the mean of 0.83. It can be demonstrated that the appearance of normally hard features is most beneficial on the medium-scale dataset. In contrast, when the larger λ of the mean caused by other α and β values is chosen, the generated tiny hard features yield limited improvement in the classification performance relative to the baseline model.

For small-scale datasets, the classification accuracy of UC Merced and NWPU 2-8 are shown in the lower half of Fig. 4(b). Due to the lack of training images, the subtle fluctuation of λ during hard feature transforming distorts the discriminative information learned by the model. In NWPU 2-8, when α and β are set at 5 and 0.1 as well as 10 and 0.1, respectively, it is reflected in Fig. 4(a) that the former λ has the possibility to be taken near 0.6. The latter λ does not have this possibility, and the discrepancy between producing normal features or not increasing the classification accuracy gap by 1.65%. In addition, UC Merced is more sensitive to the change of λ. When α and β take 1 and 0.1 as well as take 20 and 0.1, the accuracy gap reaches 12.86%. Therefore, we set α and β to be 20 and 0.1 as well as 10 and 0.1 in the two datasets, respectively, transforming the tiny hard features to boost the model performance on small-scale datasets as well as protect the rare scene information.

2) Weight ω_h in Multiscale Hard Features Joint Contrastive Loss: To investigate the effect of the weight ω_h of multilevel contrastive loss based on hard features, the following experiments are conducted: we vary the value of ω_h on each dataset to highlight the advantages of the proposed HFT.

First, the experimental results of the MLRSNet dataset are shown in Fig. 5 (top). It can be observed that the random ω_h can achieve about 93% accuracy, and when $\omega_h = 1$, the model attains the best performance of a 93.22% classification accuracy. This is because MLRSNet has abundant data, and only by adding the proportion of $\mathcal{L}_{\text{mlc}}$ based on hard features during model optimization can the potential of hard features deployed on the large dataset be developed efficiently.

Then, the accuracies in NWPU 8-2 are also demonstrated in Fig. 5 (top). We can see that this dataset is more sensitive to the change in ω_h, and the gap between the highest accuracy with ω_h and the lowest is 3.03%. However, the model has a good performance when $\omega_h < 0.1$ and $\omega_h > 0.9$; lower and higher ω_h both indicate that the generated hard features can diversify the scene information to different degrees. We finally give $\omega_h = 0.1$, and the classification accuracy of 91.35% is the highest at this time. Unlike the split of 8-2, the overall accuracy when evaluated on NWPU 2-8 is relatively stable except that $\omega_h = 0.2$ and 0.3 make a greater impact on the model classification performance. When we set ω_h as 0.1, the model in the current stage obtains the best accuracy at 76.21%.

Compared with the other three datasets, UC Merced contains the fewest training samples, and the model pretrained on a small dataset is much more fragile and highly susceptible to the synthetic scene information contained in hard features. As can be seen in Fig. 5 (bottom), the model that pays more attention to hard features learns confusing invariant representations when $\omega_h > 0.1$, which leads to the collapse of the model. Therefore, we select $\omega_h = 1$, and the classification accuracy reaches the maximum value of 77.74%.

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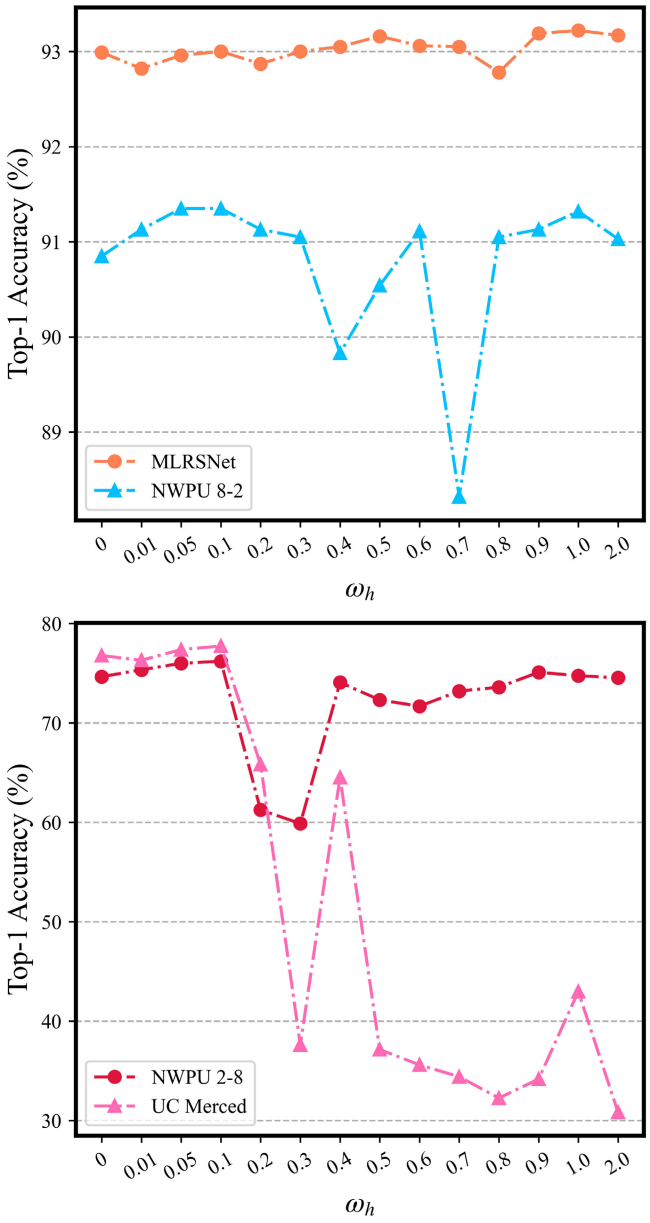


Fig. 5. Classification results with different ω_h values in each dataset.

In sum, we set $\omega_h = 0.1$ on NWPU 8-2, NWPU 2-8, and UC Merced datasets, and let $\omega_h = 1$ on the large dataset, MLRSNet, to achieve the best classification performance.

D. Ablation Study

All ablation studies in this section are conducted to show how each component contributes to MHCL, and the corresponding results are reported in Table I. Baseline denotes the model built with the basic settings described in Section IV-B. IFCL means multilevel **i**ntermediate **f**eatures **c**ontrastive **l**earning, which utilizes the multilevel contrastive loss to optimize the multilevel features; PD denotes the hard **p**ositive features **d**ecentralization operation; NC denotes hard **n**egative features **c**entralization; HFT means that we deploy the fully **h**ard **f**eatures **t**ransformation in our baseline model; finally, HFT + IFCL denotes MHCL, proposed in this article, which leverages both the HFT and multiscale hard features joint contrastive loss.

TABLE I
TOP-1 ACCURACY COMPARISON OF OUR PROPOSED MODEL WITH DIFFERENT MODULES ON FOUR DATASETS

Methods	NWPU 8-2	NWPU 2-8	UC Merced	MLRSNet
	Top-1 Accuracy (%)			
Baseline	90.85	74.65	76.78	92.99
+IFCL	91.87	77.84	80.48	93.26
+PD	91.21	75.60	73.33	93.04
+NC	91.10	75.29	75.95	92.65
+HFT	91.35	76.21	77.74	93.22
+HFT+IFCL	**92.16**	**78.80**	**81.31**	**93.68**

1) Effectiveness of IFCL: From the first and second rows of Table I, we can see that IFCL improves the performance of the model on all four datasets compared with the classic contrastive learning in the baseline. Especially in the small datasets UC Merced and NWPU 2-8, the classification accuracy is 3.7% and 3.19% higher, respectively. The features extracted by the intermediate layer of the backbone strengthen the discrimination of the model in the face of objects with different sizes in some specific scenes. Experimental results demonstrate that the multilevel contrastive loss effectively helps the model obtain better classification performance.

2) Effectiveness of PD: We first perform PD only to generate hard positive features to illustrate the effectiveness of the HFT in more detail. The corresponding results are shown in the third row of Table I, showing that PD can better supplement the inherent scene information of datasets compared with the baseline model, which boosts the accuracy on three datasets to varying degrees. However, in UC Merced with the smallest training set, PD reduces the classification result of the model by 3.45%. The main reason for this phenomenon is that the pure hard positives distort the already sparse scene discriminative information.

3) Effectiveness of NC: Similarly, we only centralize negative features to create diverse queues, while we retain the original easy positive features to verify the effectiveness of NC. From the fourth row of Table I, it can be observed that the classification accuracy increases by 0.25% and 0.64% on NWPU 8-2 and 2-8, respectively, and this number on MLRSNet and UC Merced drops slightly. The single hard negative features cannot provide extra scene information on MLRSNet but aggravate the cost of calculating loss; these reasons are just the opposite of UC Merced. In addition, we also discover that PD leads to better performance enhancement than NC on all datasets except UC Merced. The reason is that pulling the positive pairs closer helps the model learn more discriminative representations when a multilevel contrastive loss takes effect, and harder positive pairs reinforce this property better.

4) Effectiveness of HFT and Fully MHCL: We report all the results of using HFT containing both decentralization and centralization in the fifth row of Table I. The comprehensive performance advancement on all datasets compared with the baseline demonstrates that the hard positives and diverse

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TABLE II
TOP-1 ACCURACY COMPARISON OF PROPOSED MHCL WITH OTHER STATE-OF-THE-ART CONTRASTIVE LEARNING METHODS ON FOUR DATASETS. METHODS WITH * REPRESENT OUR REPRODUCED VERSION

Methods	NWPU 8-2	NWPU 2-8	UC Merced	MLRSNet
	Top-1 Accuracy (%)			
SupContrast [73]	79.11	69.50	68.81	84.51
MoCo [20]	77.94	60.39	50.95	85.75
MoCo v2 [66]	89.51	71.25	59.76	91.10
SimCLR [21]	74.89	59.60	67.02	81.89
BYOL* [22]	57.03	47.64	74.76	57.39
SimSiam* [71]	74.86	63.81	72.74	78.33
SwAV* [23]	91.18	14.38	70.70	**95.04**
PCL [74]	89.02	74.09	73.10	91.53
HCL* [77]	84.44	71.97	64.55	87.50
FT* [81]	79.22	30.82	32.74	93.50
DetCo* [75]	72.03	63.51	70.24	76.69
MHCL (ours)	**92.16**	**78.80**	**81.31**	93.68

TABLE III
TOP-5 ACCURACY COMPARISON OF PROPOSED MHCL WITH OTHER STATE-OF-THE-ART CONTRASTIVE LEARNING METHODS ON FOUR DATASETS. METHODS WITH * REPRESENT OUR REPRODUCED VERSION

Methods	NWPU 8-2	NWPU 2-8	UC Merced	MLRSNet
	Top-5 Accuracy (%)			
SupContrast [73]	97.29	93.21	94.29	94.59
MoCo [20]	93.21	87.21	88.33	93.84
MoCo v2 [66]	98.33	92.75	91.31	98.84
SimCLR [21]	97.68	88.24	93.69	92.87
BYOL* [22]	94.64	79.85	91.22	90.98
SimSiam* [71]	95.61	90.70	96.07	92.34
SwAV* [23]	**99.56**	40.11	96.43	**99.63**
PCL [74]	98.57	94.37	94.88	98.65
HCL* [77]	97.25	93.08	94.64	96.14
FT* [81]	94.59	62.31	66.12	98.61
DetCo* [75]	94.34	89.99	94.52	97.36
MHCL (ours)	99.54	**96.19**	**98.57**	99.48

queues constructed by HFT effectively enrich the inherent scene information. The comprehensive performance advancement on all datasets compared with the baseline demonstrates that the hard positives and diverse queues constructed by HFT effectively enrich the inherent scene information. In addition, the evaluation results in the last row of Table I also show that the full MHCL reaches competitive performance on all datasets. The redesigned multiscale hard features joint contrastive loss guides the model to learn invariance from multiscale easy and hard features, strengthening the discrimination of the model.

E. Comparison With Other Contrastive Learning Methods

Some recent works on scene classification [86], [87] only involved contrastive learning as a module in the model training, and their evaluation protocol is completely different from that in this article. In addition, there is also a work [88] that achieves good results, but its backbone is based on a vision transformer instead of a traditional CNN. Therefore, we compare the proposed MHCL with 11 state-of-the-art contrastive learning methods from the natural image field on four RSI datasets. The selected methods are SupContrast [73], a supervised contrastive learning method, and ten self-supervised contrastive learning methods, including the classic methods MoCo v1/v2 [20], [66], SimCLR [21], BYOL [22], and SimSiam [71]; the supervised contrastive learning method SupContrast; the clustering-based methods SwAV [23] and PCL [74]; the hard sample mining-based method HCL [77]; the data augmentation strategy-based method FT [81]; and patch contrast-based method DetCo [75]. For each comparison, we use ResNet-50 as the backbone and train 800 epochs to generate the pretrained model, and the fine-tuned settings for the downstream task are the same as those described in Section IV-B. In addition, parts of these methods have unsuitable hyperparameters for RSI datasets; thus, we adjust the appropriate hyperparameters and reproduce these methods as the final comparison.

All the above classification results are shown in Tables II and III, where it can be clearly seen that our MHCL method achieves competitive performance with the other methods on the three datasets. On the large-scale dataset MLRSNet, our top-1 and top-5 accuracies only lag behind SwAV by 1.36% and 0.15%, respectively, but outperform SwAV by a large margin on small-scale datasets, especially on UC Merced. We infer that SwAV only clusters few prototypes and learns insufficient visual representations. In contrast, MHCL can explore abundant invariant representations from small data only, resulting in significant performance gains. According to the top-1 accuracy in Table II, all comparison methods except BYOL can obtain the best results on MLRSNet, which indicates that the richer the scene information contained in the dataset, the more discriminative the invariant representations learned by the contrastive mechanism guided model. However, the top-1 accuracy of these methods on the other three small- and medium-scale datasets is unsatisfactory. Only MHCL maintains comprehensively competitive performance. Although training samples are finite, through leveraging the multilevel features from intermediate layers and transforming these to hard ones for exploring sufficient invariant visual representations, MHCL can still provide a potent upstream pretrained model for the scene classification task.

In addition, Table IV provides the time costs of different methods during training. It can be seen that HCL, FT, and DetCo with long training times do not obtain good classification performance, which is especially true for the small-scale dataset UC Merced. Although SwAV outperforms MHCL on MLRSNet, the 1.36% improvement of top-1 accuracy in exchange for nearly double more training time is subject to renegotiation. In contrast, the proposed MHCL consumes longer time than methods such as MoCo, SimCLR, and SupContrast when training on all datasets, but the significant result improvement achieved by a slightly longer training time is very valuable.

F. Analysis of Pretrained Models

Many current contrastive learning methods have been developed successively based on natural image datasets such as ImageNet-1k, which contains about 1.28 million images.

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TABLE IV

TRAINING TIME COST COMPARISON OF THE PROPOSED MHCL WITH OTHER STATE-OF-THE-ART CONTRASTIVE LEARNING METHODS ON FOUR DATASETS. METHODS WITH * REPRESENT OUR REPRODUCED VERSION

Methods	NWPU 8-2	NWPU 2-8	UC Merced	MLRSNet
	Training Time Cost (h)			
SupContrast [73]	5.39	1.94	0.75	8.70
MoCo [20]	11.01	4.15	2.41	17.98
MoCo v2 [66]	11.08	4.20	2.42	18.19
SimCLR [21]	5.43	1.97	0.76	8.73
BYOL* [22]	19.34	6.30	2.95	33.95
SimSiam* [71]	19.10	5.84	2.77	33.27
SwAV* [23]	28.72	14.40	3.48	41.86
PCL [74]	18.32	8.69	4.85	27.66
HCL* [77]	62.26	17.86	4.10	110.30
FT* [81]	54.16	13.50	3.84	90.76
DetCo* [75]	60.88	17.72	3.96	83.76
MHCL (ours)	15.85	5.40	3.27	22.92

TABLE V

TOP-1 ACCURACY COMPARISON OF DIFFERENT TYPES OF PRETRAINED MODELS ON FOUR DATASETS

Methods	NWPU 8-2	NWPU 2-8	UC Merced	MLRSNet
	Top-1 Accuracy (%)			
Random Init. [82]	24.52	18.94	28.33	25.63
ImageNet [11]	90.21	**85.75**	**95.48**	91.87
Supervised	87.95	60.91	26.19	90.88
MHCL	**92.16**	78.80	81.31	**93.68**

TABLE VI

TOP-1 ACCURACY COMPARISON OF MODELS PRETRAINED ON DIFFERENT DATASETS USING MHCL WITH 200 EPOCHS

Methods	NWPU 8-2	NWPU 2-8	UC Merced	MLRSNet
	Top-1 Accuracy (%)			
ImageNet	81.83	**78.75**	**86.83**	82.43
RSI Dataset	**86.85**	63.61	62.05	**87.97**

These methods gain good transferred performance when fine-tuning the unsupervised generated pretrained model on many downstream tasks. To investigate the superiority of the proposed MHCL method on RSI datasets, we conduct the following experiments.

Primarily, we train a ResNet-50 from scratch without loading any pretrained weights for initialization. The corresponding training parameter settings are the same as in Section IV-B, but we train 800 epochs and do not freeze the feature layers at this time. The classification results in the first row of Table V show that the random initialization of the model has a bad impact on the classification task. Next, we directly transfer the model pretrained on traditional ImageNet to the RSI scene classification task. As shown in the second row of Table V, only using the easy fine-tuning strategy can yield a good performance and achieve the best classification results on the NWPU 2-8 and UC Merced datasets. The proposed unsupervised pretraining method MHCL pales in comparison. However, the accuracy of the ImageNet pretrained model is inferior to ours on the relatively large datasets NWPU 8-2 and MLRSNet. Although the discrepancy between natural images and RSIs is obvious, the huge number of images represents rich discriminative information. For fairness, we leverage the standard supervised pretraining manner that is deployed on ImageNet to obtain a supervised pretrained model on each RSI dataset and transfer them into the scene classification task. The experimental results are shown in the third row of Table V, where it can be seen that on the basis of training images with the same order of magnitude, the proposed MHCL method generally outperforms the supervised pretraining method by a large margin, especially on the NWPU 2-8 and UC Merced datasets.

To further demonstrate the effectiveness of the proposal, MHCL is employed to pretrain on ImageNet for 200 epochs. This version of MHCL follows the settings in Section IV-B, except using $\alpha = 1$ and $\beta = 1$ to transform hard features comprehensively. After training for **175.14** h, the pretrained model achieves notable classification results, as indicated in the first row of Table VI. We compare the results with MHCL pretrained for 200 epochs on each RSI dataset. It can be seen that on the small-scale datasets NWPU 2-8 and UC Merced, the abundant visual information of ImageNet can bridge the domain gap between natural images and RSIs. However, this ability diminishes with the increase in training data. The same domain representations mined from the richer training samples of MLRSNet and NWPU 8-2 help the models achieve better performance.

According to the gap of experimental results above, we make a prospective assumption: 0.1 million RSIs are fully leveraged alone for upstream contrastive learning, and the resulting common pretrained model can achieve competitive effects beyond the ImageNet pretrained model when fine-tuning all RSI downstream tasks.

V. CONCLUSION

In this article, we introduce contrastive learning into the RSI field and proposed a novel MHCL which can provide a powerful upstream pretrained model for fine-tuning the downstream scene classification task better.

MHCL can explore sufficient invariant visual representations from the finite RSIs only. It first deploys IFCL to fully learn shallow texture and deep semantic information contained in multilevel intermediate features. Then, we propose the HFT to enrich the fixed finite scenes in small-scale RSI datasets by creating hard positive and negative features. In addition, the multiscale hard features contrastive loss is extended to mine the abundant invariant information from multiscale easy and hard features, which also boosts the discrimination of the model. When transferring the MHCL pretrained model as the initialization of the downstream scene classification task, extensive experimental results prove MHCL can achieve competitive fine-tuned performance on three popular released RSI datasets.

We hope MHCL can accelerate the pretraining research of the RSI community. In future work, the application potential of contrastive learning will continue to be exploited in the RSI remote sensing field. We will not be limited to the scene classification task, but contribute a more general upstream pretrained model for other crucial RSI tasks such as object detection, change detection, and semantic segmentation.

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Zhihao Li received the B.Eng. degree from the Xi'an University of Posts and Telecommunications, Xi'an, China, in 2019. He is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University, Xi'an.

His research interests include contrastive learning, self-supervised representation learning, and model pretraining.



Biao Hou (Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, where he is currently a Professor. His research interests include compressive sensing and synthetic aperture radar image interpretation.

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Xianpeng Guo (Student Member, IEEE) received the B.S. degree from Xidian University, Xi'an, China, in 2015, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China.

His research interests include deep learning and classification and object detection in remote-sensing images.



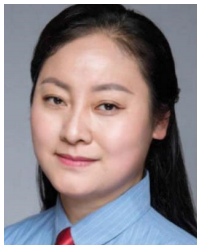
Siteng Ma received the B.S. degree from Xidian University, Xi'an, China, in 2020, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China.

His research interests include contrastive learning and scene classification and few-shot object detection in remote-sensing images.



Yanyu Cui received the B.S. degree from the North China University of Water Resources and Electric Power, Zhengzhou, China, in 2016. She is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University, Xi'an, China.

Her research interests include deep reinforcement learning and object tracking in remote sensing videos.



Shuang Wang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in circuits and systems from Xidian University, Xi'an, China, in 2000, 2003, and 2007, respectively.

She is currently a Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China, Xidian University. Her research interests include sparse representation, image processing, synthetic aperture radar (SAR) automatic target recognition, remote-sensing image captioning, and polarimetric SAR data analysis and interpretation.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Electronic Engineering, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China and the International Research Center of Intelligent Perception and Computation. His research interests include intelligent information processing, image processing, machine learning, and pattern recognition.

Dr. Jiao is a member of the IEEE Xi'an Section Execution Committee; the President of the Computational Intelligence Chapter, the IEEE Xi'an Section, and the IET Xi'an Network; the Chair of the Awards and Recognition Committee; the Vice Board Chairperson of the Chinese Association of Artificial Intelligence; a Councilor of the Chinese Institute of Electronics; a Committee Member of the Chinese Committee of Neural Networks; and an Expert of the Academic Degrees Committee of the State Council.

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